

ENSAE PARIS

MS Quantitative Methods for Economic Decisions

Fiscal Fragmentation and the Transmission of Monetary Policy in the Euro Area

ESSD - Final thesis

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 [Associated Github repository](#)

Abstract

This paper studies how fiscal fragmentation shapes the transmission of monetary policy in the euro area. We define fiscal fragmentation as the divergence in fiscal conditions across member states and measure it in real time using a novel index based on cross-country dispersion of fiscal nowcast revisions. This construction relies on incoming fiscal data rather than market prices, making it conceptually distinct from sovereign spread-based measures. Using local projections and high-frequency monetary policy surprises, we find that high-fragmentation regimes significantly amplify the contractionary effects of restrictive ECB monetary policy shocks on euro area activity.

Keywords: Nowcasting, Mixed-frequency, Fiscal fragmentation, Monetary policy transmission, Euro area

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1 Introduction

The euro area is characterized by a structural asymmetry at the core of its policy framework: while monetary policy is centralized at the European Central Bank (ECB), fiscal policy remains decentralized at the national level. This institutional asymmetry makes it important to assess whether **fiscal fragmentation**, defined in this paper as the divergence in fiscal conditions across member states, affects the transmission of common monetary policy shocks.

The ECB has increasingly addressed fragmentation risks through instruments such as the Transmission Protection Instrument (TPI), designed to preserve the smooth transmission of monetary policy in the presence of “*unwarranted and disorderly market dynamics*”¹, especially regarding sovereign spreads. However, assessing whether spread dynamics are “unwarranted” requires determining whether market tensions are justified by fiscal fundamentals. This can prove difficult: official fiscal information is released with significant delays and at low frequency, while sovereign spreads, although available in real time, are composite financial-market outcomes and do not directly isolate underlying fiscal conditions. These two limitations create a need for a real-time indicator of fragmentation based on fiscal information rather than solely on market prices.

This paper therefore asks: *how can fiscal fragmentation be measured in real time, before official fiscal data become available, and to what extent does it affect the transmission of ECB monetary policy across euro area countries?* Our contribution is twofold. First, we construct a novel real-time fiscal fragmentation index for the euro area based on country-level fiscal nowcasts, and show that it captures recent euro area fiscal developments. Second, we use this index to study whether fiscal fragmentation shapes the transmission of ECB monetary policy. Our results suggest that high-fragmentation regimes tend to amplify the effects of monetary policy shocks on economic activity. Our paper provides evidence on how a common monetary policy is transmitted in a monetary union without a fiscal union.

2 Context and conceptual framework

The euro area fragmentation is usually measured through sovereign yields or spread differentials, including price-based financial integration indicators and higher-moment measures of sovereign spreads (Kakes and Willem van den End 2023). However, spreads are an imperfect measure of fiscal stress: they also reflect liquidity premia and overall global risk sentiment. This motivates a fiscal measure of fragmentation based on incoming fiscal information rather than market prices.

Nowcasting is defined as the real-time prediction of a low-frequency macroeconomic outcome using high-frequency indicators. It is typically implemented using mixed-frequency models

¹Source: [ECB Website](#).

(Ghysels et al. 2007). Mixed-Data Sampling (MIDAS) models have largely been extended, especially to handle large information settings commonly encountered by institutions. Marcellino and Schumacher (2007) propose a MIDAS in which the high-frequency regressors are factors extracted from the data, while more recent approach rely on machine learning techniques. For instance, Babii et al. (2021) develop a sparse-group LASSO MIDAS setting to solve the same issue while improving forecasting performances. Further developments explore the use of Bayesian techniques to improve the nowcasting accuracy, as in Mogliani and Simoni (2021).

Applications to fiscal aggregates remain however sparse. Among them, Asimakopoulou et al. (2013) relies on a MIDAS model to forecast yearly fiscal aggregates for the euro area countries using quarterly indicators. In a recent paper, Cimadomo et al. (2025) relies on a mixed frequency BVAR approach to estimate the quarterly fiscal deficit of Italy using monthly macroeconomic and financial indicators. While these papers mainly aim to improve forecasting accuracy, we use nowcasting as an information-extraction tool, exploiting nowcast revisions as real-time signals of fiscal news.

Conceptually, we argue that fiscal fragmentation may amplify the transmission of monetary policy shocks, especially restrictive ones, through sovereign risk and domestic financial conditions. Sovereign yields can be decomposed into a common interest-rate component, driven by ECB policy, and a country-specific risk premium. When fiscal conditions diverge across member states, a monetary tightening may lead markets to reassess sovereign risk asymmetrically. Countries with weaker fiscal fundamentals may then experience a stronger increase in risk premia and sovereign spreads, resulting in tighter domestic financing conditions. This amplification can be reinforced by the sovereign-bank nexus. Since domestic banks hold national sovereign debt, higher sovereign yields can weaken bank balance sheets, raise funding costs, and reduce credit supply. Fiscal fragmentation may therefore turn a common monetary tightening into a more heterogeneous and contractionary transmission process across the euro area.

3 Empirical strategy

Our empirical strategy can be described in three steps. First, we nowcast each country’s current-quarter fiscal balance in a pseudo-real-time setting. Second, we define a euro area fiscal fragmentation index as the cross-country dispersion of country-level fiscal nowcast revisions. Finally, we combine this index with monetary policy surprises to test whether fiscal fragmentation affects the transmission of ECB monetary policy.

3.1 Nowcasting national fiscal balances

We estimate current-quarter fiscal aggregates using all the relevant information as it becomes available during the quarter. As recommended by Asimakopoulou et al. (2013), we separately

nowcast government revenues (R) and expenditures (E). Thus, for each country i and quarter q , the nowcasted fiscal balance B , conditional on the information available at date t , is:

$$\widehat{B}_{i,q|t} = \widehat{R}_{i,q|t} - \widehat{E}_{i,q|t}. \quad (1)$$

A key aspect of the nowcasting exercise is the reconstruction of the real-time information set. Since macro-financial and fiscal indicators are released at different dates, each vintage has a “**ragged-edge structure**”: at a given publication date, some variables may already be observed for month M , while others are only available up to $M - 1$ or earlier. We therefore build each vintage using only information available at the time, which avoids look-ahead bias and mimics the information flow available to policymakers. Figure 1 illustrates this update process.

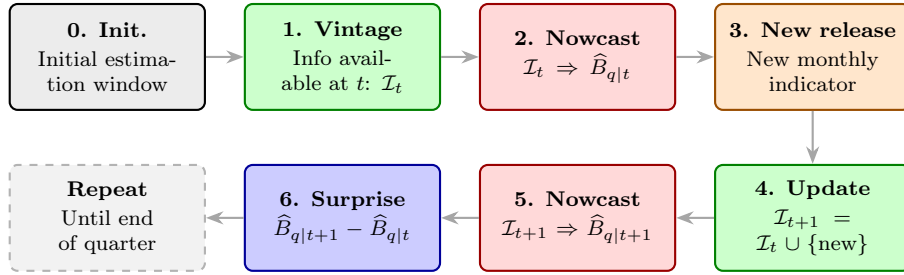


Figure 1: Pseudo-real-time nowcasting and fiscal surprise construction

Notes: At each information date, the nowcast is computed using only the information available at that date. When a new monthly release becomes available, the information set is updated and a revised nowcast is produced.

We compare two MIDAS-type nowcasting models, Factor-MIDAS (Marcellino and Schumacher 2007) and Sparse-group LASSO MIDAS (Babii et al. 2021), described in Technical Appendix C.1. All variables are seasonally adjusted using X-13 and stationarized using a sequential Augmented Dickey-Fuller procedure.

3.2 From fiscal surprises to real-time fragmentation index

The second step transforms nowcast revisions into real-time fiscal news. For each country i , the fiscal surprise at information date t , and the monthly aggregated surprise for the month m , $I_{i,m}$, are:

$$s_{i,t} = \widehat{B}_{i,q|t} - \widehat{B}_{i,q|t-1}, \quad I_{i,m} = \sum_{t \in m} s_{i,t}. \quad (2)$$

It captures the revision in the expected fiscal balance induced by newly released information.

The euro area fragmentation index is then defined as the GDP-weighted cross-country dispersion of country-level fiscal news. The underlying intuition is that cross-country divergence in fiscal trajectories may alter the aggregate transmission of ECB monetary policy, as argued

in the [conceptual framework](#):

$$Frag_m^{raw} = \sqrt{\sum_{i=1}^4 w_i (I_{i,m} - \bar{I}_m)^2}, \quad (3)$$

where $\bar{I}_m = \sum_{i=1}^4 w_i I_{i,m}$. For interpretation, we log-transform and standardize the raw index:

$$Frag_m = 100 + 10 \times \frac{\ell_m - \mu_\ell}{\sigma_\ell} \quad ; \quad \ell_m = \log(Frag_m^{raw}), \quad (4)$$

where μ_ℓ and σ_ℓ denote the long-run mean and standard deviation of ℓ_m . Values above 100 indicate higher-than-usual fiscal fragmentation. It therefore aims to capture the underlying fiscal dimension of fragmentation: not whether markets already price sovereign risk differently, but whether incoming fiscal information signals diverging fiscal trajectories across member states.

3.3 Fiscal fragmentation and monetary policy transmission

Finally, we assess whether fiscal fragmentation affects the transmission of ECB monetary policy. Monetary policy shocks are identified using high-frequency monetary policy surprises, described in Technical Appendix [C.2](#). We first estimate:

$$\Delta Y_t = \alpha + \beta MPS_t + \gamma (MPS_t \times Frag_t) + \delta Frag_t + X_t' \theta + \varepsilon_t, \quad (5)$$

where Y_t is the euro area production factor, $Frag_t$ is the standardized fiscal fragmentation index, and X_t is a vector of macro-financial controls. The coefficient of interest, γ , measures whether monetary policy transmission depends on fiscal fragmentation. Under the convention that positive monetary policy surprises are restrictive, we expect $\gamma < 0$, as it amplifies the negative effect of the monetary policy shock on production.

We then estimate local projections à la [Jordà \(2005\)](#):

$$Y_{t+h} - Y_{t-1} = \alpha_h + \beta_h MPS_t + \gamma_h (MPS_t \times Frag_t) + \delta_h Frag_t + X_t' \theta_h + \varepsilon_{t+h}^{(h)}. \quad (6)$$

The conditional response to a monetary policy shock is therefore:

$$\frac{\partial(Y_{t+h} - Y_{t-1})}{\partial MPS_t} = \beta_h + \gamma_h Frag_t. \quad (7)$$

This allows us to compare responses under low, average and high fiscal fragmentation. Standard errors are Newey-West corrected.

A potential concern is whether $Frag_t$ is exogenous to monetary policy surprises. Two features of our construction address this. First, MPS_t is identified from asset price movements

in a narrow window around ECB meetings, isolating the non-anticipated component of policy decisions by construction. Second, $Frag_t$ is built from nowcast revisions driven by incoming fiscal data releases, not from market prices, and is therefore unlikely to embed anticipations of future monetary policy, unlike sovereign spreads.

4 Data

We construct the fiscal fragmentation index for Germany, France, Italy and Spain over 2010–2025. The nowcasting target is the quarterly general government balance, computed from official S13 government revenues and expenditures. Predictors include monthly cash fiscal data, manually collected from national treasury publications, alongside macro-financial indicators from DBnomics and Eurostat, covering activity, labor markets, prices, and confidence. Each variable is included on grounds of economic relevance and consistency with the existing literature. The full list of variables, sources and release schedules is reported in Appendix A2 and A3.

Monetary policy surprises are constructed using Altavilla et al. (2025). Dependent variables and controls are taken from EA-MD (Barigozzi and Lissona 2026). For each macroeconomic block, we extract the first principal component of the available indicators, yielding synthetic factors for production, labor markets, prices, financial conditions and confidence.

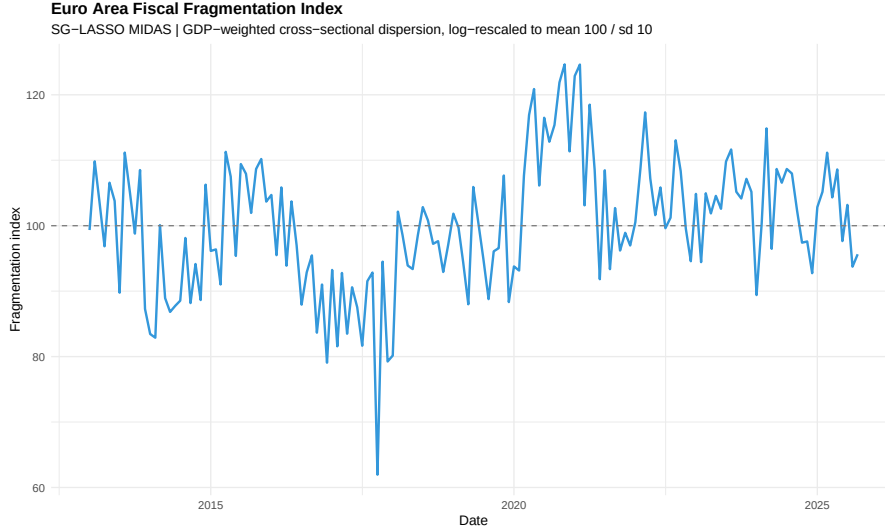
5 Main results

Using data for France, Germany, Italy and Spain over 2010–2025, we implement the nowcasting strategy described in Section 3.1 with an expanding window. Predictive performance is reported in Appendix Table A4: our MIDAS-type models do not systematically outperform simple AR benchmarks in terms of point forecast accuracy, but their forecasting performance remains comparable. Among them, SG-LASSO performs best on average and is retained as the baseline specification.

Figure 2 presents the real-time euro area fiscal fragmentation index. Fiscal fragmentation increased sharply after 2020 and remained above its historical average. This pattern is consistent with the heterogeneous fiscal responses, both in size, and composition, to the Covid-19 shock documented by the ECB (Haroutunian et al. 2021). The fact that our index captures this well-documented episode of heterogeneous fiscal responses supports its relevance as a real-time measure of fiscal fragmentation.

Table 1 reports the static interaction regression. The interaction coefficient is negative and statistically significant: a one-standard deviation increase in fiscal fragmentation makes the marginal effect of a restrictive monetary policy surprise more negative by about 2.17 units of the production factor. This result is robust to alternative dependent variables, to using Factor-

Figure 2: Real-time euro area fiscal fragmentation index



Notes: Transformed real-time euro area fiscal fragmentation index as in Equation (4). Values above 100 indicate a level of fiscal fragmentation higher than its historical average, while values below 100 indicate lower-than-average fragmentation. Model used: SG-LASSO.

MIDAS instead of SG-LASSO nowcasts, and to replacing GDP-weighted country factors with direct euro area factors, although the latter specification is weaker (Appendix Tables A5, A6 and A7). Decomposing monetary policy surprises by sign shows that the amplification effect is concentrated among restrictive surprises, consistent with the sovereign-risk channel (Appendix Table A8).

Table 1: Static euro area regression: GDP-weighted country aggregates

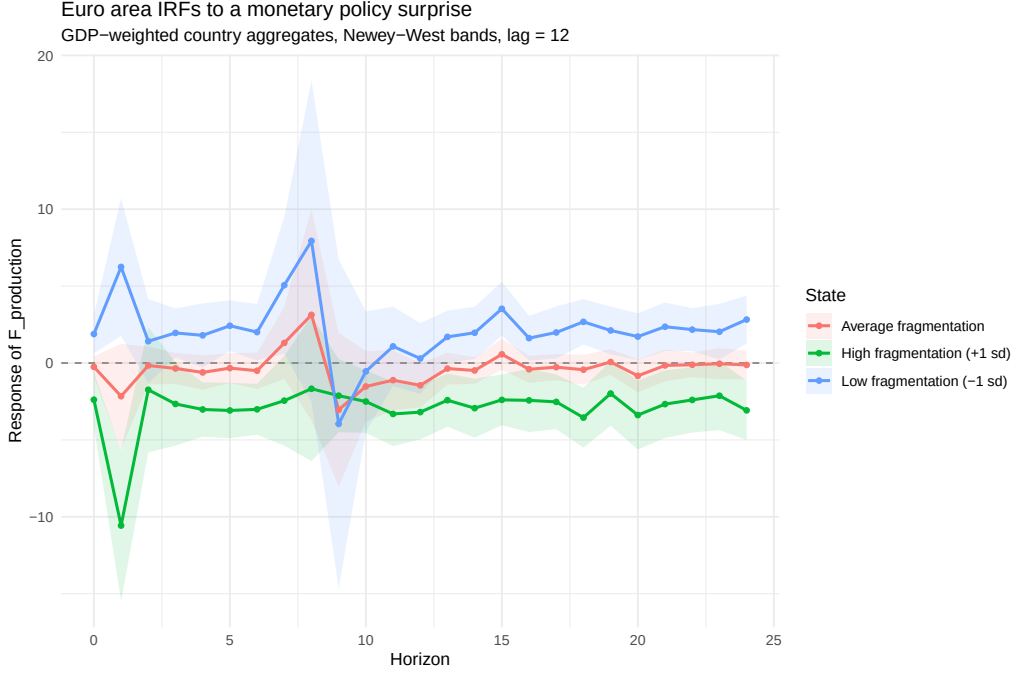
	ΔY_t
$MPS_t \times Frag_t$	-2.1675^{***} (0.5871)
Labor	-3.6348^{**} (1.8223)
Financials	5.6964^{***} (0.9369)
Turnover	-1.9349^{***} (0.5173)
Confidence	-2.2481^{***} (0.4846)
Observations	137
R^2	0.8157

Notes: Newey-West standard errors in parentheses, computed with 24 lags. ***, ** and * denote significance at the 1%, 5% and 10% levels. $Frag_t$ is standardized. α, β, δ and Prices are not significant and thus not reported.

The dynamic evidence from local projections confirms this result. Figure 3 reports conditional impulse responses under low, average and high fiscal fragmentation regimes, following Equation (7). Under high fragmentation, a restrictive monetary policy shock generates a clearly more negative response of the production factor. This confirms the main empirical result of the paper: fiscal fragmentation amplifies the transmission of restrictive ECB monetary policy

shocks to euro area activity. Horizon-specific interaction coefficients, reported in Appendix A2, are negative and statistically significant, consistent with the rest of our results.

Figure 3: Conditional response of euro area activity to a monetary policy shock



Notes: This figure reports conditional impulse responses from euro area local projections using GDP-weighted country aggregates. The dependent variable is the euro area production factor. The responses are computed as $\beta_h + \gamma_h Frag$, where $Frag$ is evaluated at low fragmentation (-1 standard deviation), average fragmentation, and high fragmentation ($+1$ standard deviation). Confidence bands are Newey-West corrected with 12 lags.

6 Conclusion

This paper constructs a real-time fiscal fragmentation index for the four largest euro area economies using country-level fiscal nowcast revisions. Unlike sovereign spreads, the index is based on incoming fiscal information rather than market prices. We document a sharp rise in fiscal fragmentation after 2020 and show that high-fragmentation regimes amplify the contractionary effects of restrictive ECB monetary policy shocks on euro area activity. These findings have direct policy implications for the design of tools such as the TPI, which relies on real-time assessments of fiscal fundamentals to assess whether market dynamics are “unwarranted”.

Several limitations remain. First, the exercise is pseudo-real-time: we reproduce the original publication calendar and ragged-edge structure, but fully unrevised vintage data are not available for all indicators. Second, nowcast revisions may contain both genuine fiscal news and model noise. Third, the index is built on four countries only and should be compared more systematically with market-based fragmentation measures such as sovereign spreads. However, a lack of access to market data made this comparison impossible. Future work could extend the country coverage, use full real-time vintages, and explore alternative nowcasting specifications.

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A Data details

Table A2: Sources for the Nowcasting aggregates

Series	France	Germany	Italy	Spain
Total revenue & Total expenditures	Eurostat, Quarterly data, Seasonally and Calendar adjusted, General government			
10-year yields	DBnomics - BDF <i>BDF/FM/M.FR.EUR.FR2.BB.FR10YT_RR.YLD</i>			
Payroll Mass	DBnomics - Eurostat, Compensation of Employees <i>Eurostat/NAMQ_10_GDP/Q.CP_MEUR.NSA.D1</i>			
Hours worked	DBnomics - Eurostat, Index of hours worked <i>Eurostat/namq10a10e/Q.I15HW.TOTAL.SCA.EMPDC</i>			
Hourly earnings	DBnomics - Eurostat, Hourly Labor cost index (2020 = 100) <i>Eurostat/lcici_r2q/Q.SCA.I20.B - S.D11</i>			
IP	Banque de France, Monthly data, Seasonally and calendar adjusted, Industrial Production Index - construction excluded (2021 = 100)			
Business turnover	Eurostat, Monthly, Seasonally and calendar adjusted, Turnover in industry - total (2021 = 100)			
ESI	DBnomics - Eurostat, Economic Sentiment Indicator <i>Eurostat/ei_bssi_m_r2/M.BS-ESI-I.SA</i>			
Retail trade	Banque de France, Monthly data, Seasonally and calendar adjusted, Retail trade turnover index deflated (2021 = 100)			
Private consumption	DBnomics - Eurostat, Final consumption expenditure of household in current prices <i>Eurostat/namq_10_gdp/Q.CP_MEUR.SCA.P31_S14</i>			
HICPX	DBnomics - Eurostat, HICP index excluding energy <i>Eurostat/prc_hicp_manr/M.RCHA_TOT_X_NRG</i>			
Energy prices	DBnomics - Eurostat, HICP energy component (2015=100) <i>Eurostat/prc_hicp_midx/M.I15.NRG</i>			
Stock market returns	Yahoo finance, Monthly log returns			
Debt-to-GDP	Eurostat, Quarterly, Government consolidated gross debt expressed in percentage of GDP - central government			
Unemployment rate and inflows	DBnomics - OECD, Monthly, Seasonally and calendar adjusted, unemployment rate			
Share of LT unemployed	DBnomics, Eurostat, Long term unemployment (Percentage of unemployed) <i>Eurostat/UNE_LTU_Q/Q.LTU.Y15-74.PC_UNE.SA.T</i>			
Geopolitical risk index	Newspaper based index of geopolitical risk (Caldara and Iacoviello GPR index)			
Monthly cash data	SMB	BMF	Mensual report from MEF	MEF

The sources for the publication delays of all Euro indicators, taken from Eurostat, can be found in the [release calendar for Euro indicators](#).

Table A3: Publication delays used in the analysis

Series	France	Germany	Italy	Spain
Total revenue & Total expenditures	Flash estimate 85 days after the end of the quarter. Updated publication 112 days after the end of the quarter.			
10-year yields	Available daily, no publication delay			
Payroll Mass	Flash estimate 45 days after end of each quarter. Updated estimates published around $t + 65$ and $t + 110$			
Hours worked	Flash estimate 45 days after end of each quarter. Updated estimates published around $t + 65$ and $t + 110$			
Hourly earnings	75 days after the end of the reference quarter			
IP	75 days after the end of the reference month (15 of M+2)			
Business turnover	On the first week 2 months after the end of the reference month			
ESI	Last thursday of current month			
Retail trade	First thursday M+2			
Private consumption	Estimates published around $t+65$ and $t+110$			
HICP	Flash on the last working day of current month ; Definitive around the 15th of M+1			
Energy prices	Flash on the last working day of current month ; Definitive around the 15th of M+1			
Stock market returns	Available daily, no publication delay			
Debt-to-GDP	Flash estimate 45 days after end of each quarter. Updated estimates published around $t + 65$ and $t + 110$			
Unemployment rate and inflows	Estimate $t + 15$, publication $t + 40$			
Share of LT unemployed	75 days after the end of the reference quarter			
Geopolitical risk index	Ex-post index, no publication delay			
Monthly cash data	Middle of the following month	Middle of the following month	End of the following month	Middle of the following month

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B Additional Results

B.1 Nowcasting performance

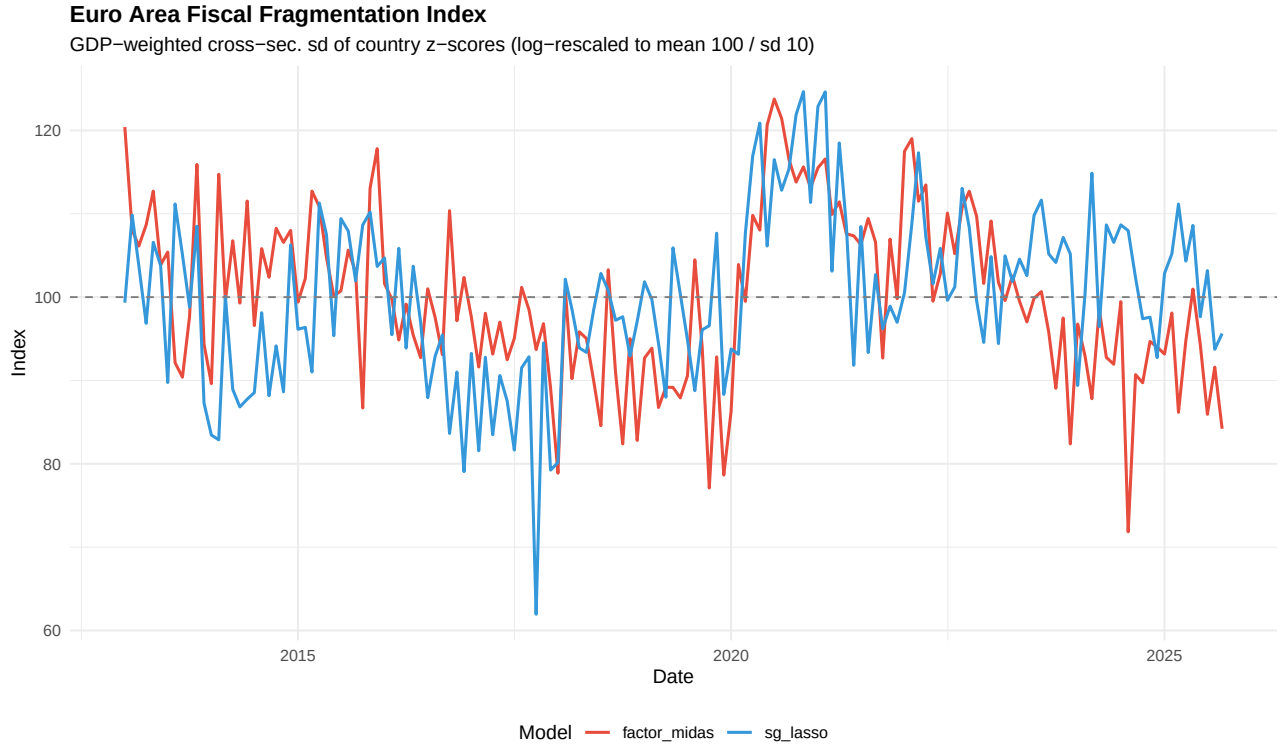
Table A4: Nowcasting performance by model and country

Model	Country	RMSE	MAE
AR(1)	Average	6831.44	4013.20
	France	7770.13	3955.38
	Germany	8202.16	5364.20
	Italy	6215.75	3637.57
	Spain	5137.69	3095.65
Factor-MIDAS	Average	8532.27	4900.14
	France	7288.41	4085.79
	Germany	11074.18	7278.76
	Italy	7662.07	4304.98
	Spain	8104.42	3931.02
Random walk	Average	9297.25	5415.71
	France	9558.99	4889.86
	Germany	11866.13	7440.50
	Italy	8321.78	5164.70
	Spain	7442.10	4167.79
SG-LASSO MIDAS	Average	7825.98	4472.97
	France	7327.75	3825.98
	Germany	9152.79	5906.40
	Italy	7438.87	4213.56
	Spain	7384.49	3945.93

Notes: RMSE denotes the root mean squared error, and MAE denotes the mean absolute error. The reported values are averages of the RMSE and MAE computed at each nowcast date, for each fiscal block. The maximum number of nowcast observations is 979, although the effective number of observations may be lower for Factor-MIDAS and SG-LASSO MIDAS because missing values can occur at the beginning of the pseudo-real-time sample. Simple time-series benchmarks perform relatively well, which reflects the persistence of fiscal variables and the difficulty of improving forecasts of stationary fiscal series. However, the objective of the exercise is to extract infra-monthly fiscal surprises from the flow of newly released macro-financial information, which is by definition impossible with AR specifications. This motivates the use of MIDAS-type models for the construction of the real-time fiscal fragmentation index. ([return to main text](#)).

B.2 Robustness checks

Figure A1: Real-time euro area fiscal fragmentation index - Factor-MIDAS vs. SG-LASSO



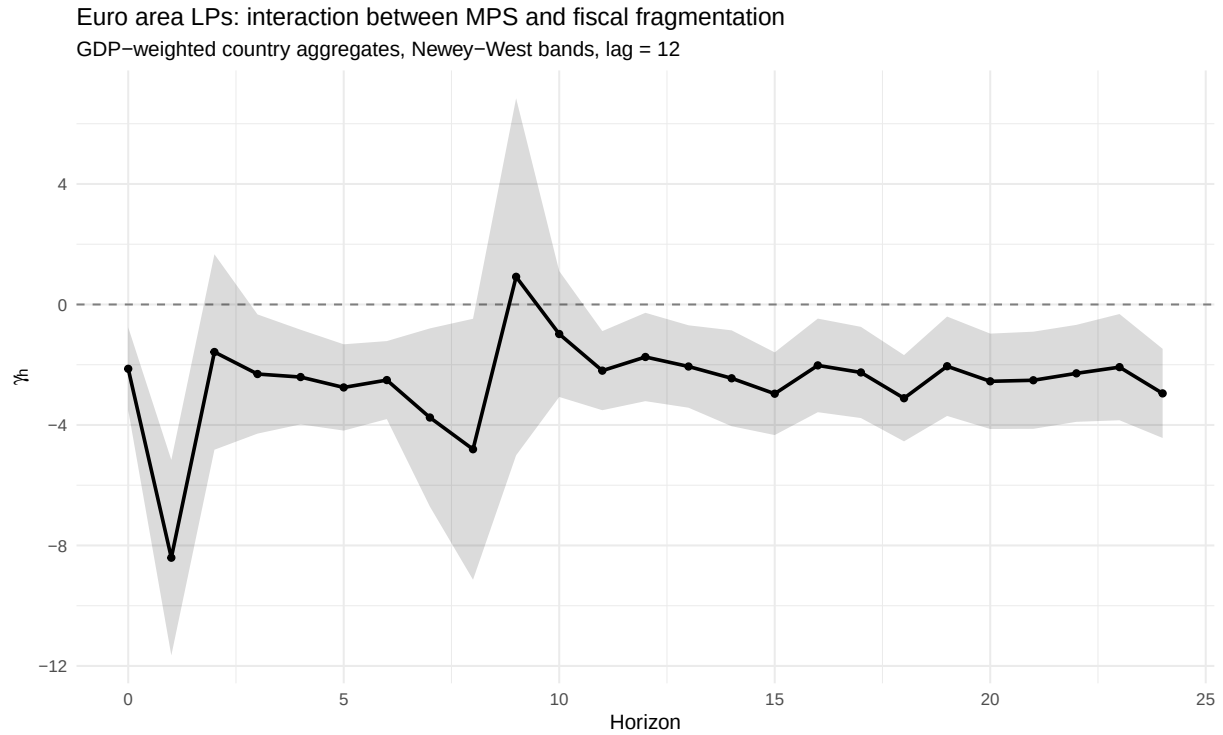
Notes: Transformed real-time euro area fiscal fragmentation index as in Equation (4). Values above 100 indicate a level of fiscal fragmentation higher than its historical average, while values below 100 indicate lower-than-average fragmentation. In red (resp. blue) is the index using Factor-MIDAS (resp. SG-LASSO) nowcast surprises.

Table A5: Robustness: static interaction coefficient across alternative dependent variables

Dependent variable	γ Estimate	Newey-West s.e.	p -value	R^2
Financials	-1.6948***	0.2553	< 0.001	0.7567
Turnover	-2.5019***	0.4494	< 0.001	0.8190
Industrial production	-2.1675***	0.5871	< 0.001	0.8157
Prices	-0.2515***	0.0777	0.0015	0.1047
Confidence	-0.6951***	0.2623	0.0090	0.6609
Labor	-0.0706*	0.0375	0.0619	0.1624

Notes: This table reports the coefficient on $MPS_t \times Frag_t$ from static regressions using alternative EA-MD factors as dependent variables. Newey-West standard errors are reported in the third column, using 24 lags. In each regression, the dependent variable is the monthly change in the corresponding factor, and the remaining EA-MD factors are included as controls. $Frag_t$ is the standardized fiscal fragmentation index. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels. ([return to main text](#))

Figure A2: euro area local projections: interaction between monetary policy and fiscal fragmentation



Notes: This figure reports the horizon-specific interaction coefficients γ_h from euro area local projections using GDP-weighted country aggregates. The coefficient γ_h measures whether fiscal fragmentation modifies the effect of monetary policy surprises on euro area activity at horizon h . The dependent variable is the euro area production factor. The fiscal fragmentation index is standardized, so γ_h can be interpreted as the change in the marginal effect of a monetary policy surprise associated with a one-standard deviation increase in fiscal fragmentation. Shaded areas correspond to Newey–West confidence bands with 12 lags.

Table A6: Static euro area regression: GDP-weighted country aggregates, Factor-MIDAS fragmentation index

Main variables					
	Intercept	MPS_t	$Frag_t$	$MPS_t \times Frag_t$	
Estimate	0.2300	-0.2876	0.3643*	−1.8009**	
Newey-West s.e.	(0.2095)	(0.4318)	(0.1999)	(0.8790)	
p -value	0.2742	0.5065	0.0706	0.0424	
Controls					
	Labor	Financials	Turnover	Prices	Confidence
Estimate	-3.0729*	5.8242***	-2.0507***	-0.3323	-2.3694***
Newey-West s.e.	(1.6887)	(1.0244)	(0.5484)	(0.3040)	(0.4440)
p -value	0.0710	< 0.001	< 0.001	0.2763	< 0.001
Observations	137				
R^2	0.8053				

Notes: Newey-West standard errors are reported in parentheses. The covariance matrix is Newey-West corrected with 24 lags to account for residual autocorrelation in the static regression. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels. The dependent variable is the change in the euro area production factor. $Frag_t$ is the **standardized** fiscal fragmentation index obtained from the Factor-MIDAS nowcasts, so that coefficients can be interpreted in terms of one-standard deviation changes in fragmentation.

Table A7: Static euro area regression: direct euro area EA-MD factors

	ΔY_t	
	SG-LASSO index	Factor-MIDAS index
Intercept	0.2831* (0.1456)	0.1808 (0.1421)
MPS_t	-0.7710*** (0.2812)	-1.0992*** (0.3912)
$Frag_t$	-0.3651** (0.1647)	0.1256 (0.1385)
$MPS_t \times Frag_t$	-1.0228* (0.5376)	0.0959 (0.6739)
Labor	-2.6746** (1.2693)	-1.8164* (1.0787)
Financials	-0.4251*** (0.1515)	-0.4436*** (0.1604)
Turnover	0.9670*** (0.2563)	0.9333*** (0.2497)
Prices	-1.4623*** (0.4982)	-1.3361*** (0.4719)
Confidence	-0.8449* (0.4812)	-0.9839** (0.4620)
Observations	137	137
R^2	0.5828	0.5681

Notes: Newey-West standard errors are reported in parentheses. The covariance matrix is Newey-West corrected with 14 lags to account for residual autocorrelation in the static regression. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels. The dependent variable is the change in the direct euro area production factor from the EA-MD database. $Frag_t$ is the standardized real-time fiscal fragmentation index. The first column uses the SG-LASSO-based fragmentation index, while the second column uses the Factor-MIDAS-based fragmentation index. The highlighted interaction coefficient is the parameter of interest.

Table A8: Static euro area regression with sign-decomposed monetary policy surprises

	ΔY_t	
	GDP-weighted factors	Direct EA factors
Intercept	0.2011 (0.1953)	0.3639 (0.2324)
MPS_t^+	0.1631 (0.2896)	-0.9191* (0.5309)
MPS_t^-	-1.0489*** (0.3109)	-0.1798 (1.2058)
$Frag_t$	0.2291 (0.1538)	-0.3068 (0.2474)
$MPS_t^+ \times Frag_t$	-2.5223*** (0.7110)	-1.1638 (0.7576)
$MPS_t^- \times Frag_t$	-0.5176 (1.0232)	-0.1456 (1.0437)
Labor	-3.5845* (1.8518)	-2.7420** (1.2098)
Financials	5.7113*** (0.9364)	-0.4194*** (0.1553)
Turnover	-1.9434*** (0.5199)	0.9718*** (0.2518)
Prices	-0.3865 (0.2865)	-1.4397*** (0.5281)
Confidence	-2.2363*** (0.5189)	-0.8099 (0.5396)
Observations	137	137
R^2	0.8157	0.5828

Notes: Newey-West standard errors are reported in parentheses. The dependent variable is the change in the euro area production factor. $MPS_t^+ = MPS_t \mathbb{1}(MPS_t > 0)$ and $MPS_t^- = MPS_t \mathbb{1}(MPS_t < 0)$ denote positive and negative monetary policy surprises, respectively. $Frag_t$ is the standardized SG-LASSO real-time fiscal fragmentation index. The interaction terms are highlighted as the parameters of interest. ***, ** and * denote statistical significance at the 1%, 5% and 10% levels.

C Technical Appendix

C.1 Nowcasting models

We provide additional details on the two nowcasting models used to construct the real-time fiscal surprise index. For each country i and fiscal block $b \in \{\text{revenues, expenditures}\}$, the objective is to nowcast the current-quarter fiscal variable $b_{i,q}$ using monthly macro-financial information. The fiscal balance nowcast is then obtained as:

$$\hat{B}_{i,q|t} = \hat{R}_{i,q|t} - \hat{E}_{i,q|t},$$

where

$$\hat{R}_{i,q|t}$$

and

$$\hat{E}_{i,q|t}$$

denote respectively the nowcasted revenues and expenditures at information date t .

C.1.1 Factor-MIDAS

Following [Marcellino and Schumacher \(2007\)](#), we summarize a large set of monthly indicators into a small number of common factors, which are then used in a mixed-frequency equation. Let X_t denote the vector of monthly predictors available at month t . We assume that these predictors can be summarized by a small number of latent factors:

$$X_t = \Lambda F_t + u_t,$$

where F_t is a vector of common factors, Λ the loading matrix and u_t an idiosyncratic component. Factors are estimated by principal components, after standardizing the monthly predictors (since this is done each time we have a new information in our pseudo-real time setting, the means and standard deviations are computed with data up to t , so no forward bias is introduced). The number of factors is selected using a Bai-Ng criterion. It selects the number of factors by minimizing a criterion that balances goodness-of-fit against a penalty increasing in the number of factors, to avoid overfitting.

$$\hat{k} = \arg \min_{k \in \{0, \dots, k_{\max}\}} \left\{ \log \left[\frac{1}{NT} \sum_{t=1}^T \sum_{j=1}^N \left(X_{tj} - \hat{X}_{tj}^{(k)} \right)^2 \right] + k \frac{N+T}{NT} \log(\min(N, T)) \right\}.$$

In our analysis, we set $k_{\max} = 4$. If $\hat{k} = 0$, we set the number of factors to 1 to preserve the Factor-MIDAS structure and ensure at least one common component is extracted. [Table A9](#) shows that the selected number of factors varies across countries and fiscal blocks, with average values ranging from 1.06 to 2.93.

Table A9: Number of factors selected in the Factor-MIDAS model

Country	Fiscal block	Mean	Median	Min.	Max.
France	Expenditures	2.01	2	1	4
France	Revenues	2.26	2	1	4
Germany	Expenditures	2.88	3	1	4
Germany	Revenues	2.52	2	1	4
Italy	Expenditures	1.06	1	1	4
Italy	Revenues	2.43	2	1	4
Spain	Expenditures	2.93	2	1	4
Spain	Revenues	1.52	1	1	4

Notes: This table reports summary statistics for the number of factors selected by the Bai-Ng PC_{p2} criterion in the Factor-MIDAS model. The selection is performed separately for each country, fiscal block and pseudo-real-time estimation window. The maximum number of factors is set to $k_{\max} = 4$. If the criterion selects zero factors, one factor is retained by default to preserve the Factor-MIDAS structure.

Once the factors are estimated, the quarterly fiscal variable (accrual expenditures or revenues) is projected on its own lag and on current and lagged monthly factors:

$$b_q = \alpha + \rho b_{q-1} + \sum_{r=1}^K \sum_{\ell=0}^{L_F} \beta_{r,\ell} F_{r,m(q)-\ell} + \varepsilon_q,$$

where $m(q)$ denotes the last month of quarter q , K is the number of selected factors, and L_F is the number of monthly factor lags. In the baseline specification, we set:

$$k_{\max} = 4, \quad L_F = 2.$$

The Factor-MIDAS nowcast is obtained by applying the estimated regression coefficients to the most recent factor values available at the nowcast date. The resulting nowcast is then transformed back to the original scale of the fiscal variable.

C.1.2 Sparse-group LASSO MIDAS

Our implementation largely follows [Babii et al. \(2021\)](#) and relies on the R package developed by the authors. Unlike the Factor-MIDAS model, this approach does not first compress the predictors into principal components. Instead, it directly uses lagged monthly predictors and relies on regularization to select the relevant information.

For each quarterly observation q , let $X_{j,m(q)-\ell}$ denote monthly predictor j observed at monthly lag ℓ . The unrestricted MIDAS regression can be written as:

$$b_q = \alpha + \sum_{p=1}^{P_y} \rho_p b_{q-p} + \sum_{j=1}^N \sum_{\ell=0}^{L_X} \beta_{j,\ell} X_{j,m(q)-\ell} + \varepsilon_q.$$

In the baseline implementation, we set:

$$P_y = 1, \quad L_X = 2.$$

The model therefore includes one quarterly lag of the fiscal target and, for each monthly predictor, the current monthly value and the previous two monthly lags.

However, including each monthly lag separately can quickly increase the number of parameters. To keep the model parsimonious, each block of monthly lags is summarized using Legendre polynomial transformations. For predictor j , define the vector of monthly lags:

$$Z_{j,q} = (X_{j,m(q)}, X_{j,m(q)-1}, \dots, X_{j,m(q)-L_X})'.$$

This vector is then transformed as:

$$\tilde{Z}_{j,q} = Z_{j,q}' W,$$

where W is a matrix of Legendre polynomial weights. Each column of W corresponds to one Legendre polynomial evaluated over the lag positions. Hence, each element of $\tilde{Z}_{j,q}$ is a weighted average of the monthly lags of predictor j . If the Legendre degree is d , the transformation produces $d+1$ regressors for predictor j , instead of L_X+1 unrestricted monthly lag coefficients.

In our baseline specification, we set $d = 0$.: each predictor-specific lag block is compressed

into a single transformed regressor. With $L_X = 2$, the model starts from:

$$(X_{j,m(q)}, X_{j,m(q)-1}, X_{j,m(q)-2}),$$

and summarizes these three monthly observations into one low-dimensional component. This is a deliberately parsimonious choice, motivated by the relatively short quarterly estimation sample and the pseudo-real-time nature of the exercise.

The transformed regression can then be written compactly as:

$$b_q = \alpha + \tilde{X}_q' \beta + \varepsilon_q,$$

where $\tilde{X}_q$ contains the quarterly target lag and the Legendre-transformed MIDAS blocks of all monthly predictors.

The sparse-group LASSO estimator solves:

$$\hat{\beta} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_{q=1}^T (b_q - \alpha - \tilde{X}_q' \beta)^2 + 2\lambda \Omega(\beta) \right\},$$

with penalty:

$$\Omega(\beta) = \gamma \|\beta\|_1 + (1 - \gamma) \sum_{g \in \mathcal{G}} \|\beta_g\|_2.$$

The penalty has two components. The first term, $\|\beta\|_1$, is the standard LASSO penalty and induces sparsity at the coefficient level by shrinking some coefficients exactly to zero. The second term, $\sum_{g \in \mathcal{G}} \|\beta_g\|_2$, is the group-LASSO penalty and induces sparsity at the group level. In our setting, a group corresponds to one monthly predictor and its transformed MIDAS components. Thus, selecting a group means that the corresponding predictor is retained in the nowcasting equation, while discarding a group means that this predictor is excluded. Since our baseline uses $d = 0$, each predictor group contains only one transformed component, so group selection is equivalent to selecting or excluding one predictor. The autoregressive lag of the fiscal target is included but left unpenalized.

The parameter λ controls the overall strength of regularization. A higher value of λ implies stronger penalization, more shrinkage, and therefore a more parsimonious specification with fewer selected predictors. Conversely, a lower value of λ allows more predictor groups to enter the model. The parameter γ controls the relative weight given to coefficient-level and group-level sparsity. We fix $\gamma = 0.5$, which corresponds to a neutral compromise between pure LASSO ($\gamma = 1$) and pure group LASSO ($\gamma = 0$). This choice is motivated by the pseudo-real-time nature of the exercise. Although the full sample eventually contains enough observations, the early estimation windows are short and the model is re-estimated at each information date. Jointly tuning both γ and λ would therefore add an additional layer of instability and computational burden.

We instead fix γ and select only the overall regularization strength λ by time-series cross-validation. The number of folds is set as:

$$K = \min \left(10, \left\lfloor \frac{T}{3} \right\rfloor \right),$$

where T is the number of available estimation observations. The validation block length is set to 2. This cross-validation scheme accounts for the time-series nature of the data by using contiguous validation blocks rather than randomly splitting observations.

All regressors and the target variable are standardized before estimation. The centering and scaling parameters are computed only on the estimation sample and then applied to the nowcast row, avoiding any look-ahead bias. The prediction is first obtained on the standardized scale and then transformed back to the original scale of the fiscal variable.

Table A10: Variable selection in the SG-LASSO MIDAS model

Country	Fiscal block	Selected groups		λ	
		Mean	Median	Mean	Median
France	Expenditures	3.76	2	0.316	0.334
France	Revenues	7.67	8	0.114	0.057
Germany	Expenditures	4.07	3	0.280	0.263
Germany	Revenues	5.55	6	0.251	0.088
Italy	Expenditures	5.25	2	0.176	0.158
Italy	Revenues	4.05	2	0.280	0.270
Spain	Expenditures	5.96	4	0.205	0.123
Spain	Revenues	5.33	5	0.100	0.092

Notes: This table reports the average and median number of predictor groups selected by the SG-LASSO MIDAS model across pseudo-real-time estimation windows, together with the average and median values of the regularization parameter λ . Higher values of λ imply stronger penalization and therefore more parsimonious specifications. The selection is performed separately for each country and fiscal block.

Because the nowcasting models are re-estimated at each pseudo-real-time date, coefficient-level inference is not the main object of interest. Factor-MIDAS coefficients vary across vintages, and the extracted factors themselves may change as the information set evolves. For SG-LASSO MIDAS, standard coefficient-level inference is also not directly applicable because the estimates are obtained after regularized variable selection. We therefore assess the nowcasting models using predictive performance metrics and model-selection diagnostics rather than interpreting individual coefficients.

C.2 Monetary policy surprises

Monetary policy surprises are used to measure monetary policy shocks, defined as the non-anticipated part of a given monetary event. In the literature, they are typically computed as the variation of asset prices, typically futures on interest rates, in a narrow window around a

central bank meeting (ECB in our case). Formally, [Gürkaynak et al. \(2005\)](#) define MPS as:

$$mps_t^{(h)} = p_t^{(h)} - p_{t-\Delta t}^{(h)} = \underbrace{\left[F_t^{market}(i_{t+h}) - F_{t-\Delta t}^{market}(i_{t+h}) \right]}_{\text{Expectation revision}} + \underbrace{\left[\zeta_t^{(h)} - \zeta_{t-\Delta t}^{(h)} \right]}_{\approx 0}$$

with:

- $p_t^{(h)}$: price of futures expiring at $t + h$
- $F_t^{market}(i_{t+h})$: market forecast of the policy path between t and $t + h$
- $\zeta_t^{(h)}$: risk premium.

In their seminal paper, [Gürkaynak et al. \(2005\)](#) decompose MPS into two distinct factors:

- A **path** factor, which reflects revisions to the expected future path of monetary policy
- A **target** factor, which captures surprises in the current policy rates

MPS are usually computed as the most important latent factors driving the price variation over a wide range of asset prices. They are typically retrieved using a PCA.

In our empirical application, we rely on the [Gürkaynak et al. \(2005\)](#) decomposition and compute the Target/Path factor following the approach of [Herbert et al. \(2025\)](#). In particular:

- The Target factor is computed as the change in 1-month OIS rates around all events recorded in the database
- The Path factor is computed as the change in 1-year OIS rates, orthogonalized to changes in 1-month OIS rates, as it conveys information related to future monetary policy decisions
- Both factors are scaled such that their standard deviation equals 1

The monetary policy surprises we use are then defined as:

$$\boxed{MPS_t = Path_t + Target_t}$$

In terms of interpretation, a positive monetary policy surprise implies that there was either an upward repricing of future contracts, which embed expectations about the future stance of policy rates. Hence, it can be interpreted as a contractionary monetary policy shock.

Finally, we build a measure of monetary policy shocks over a given month as the sum of all surprises that occurred during this month. This is standard in the literature that studies the transmission of monetary policy to macroeconomic aggregates, such as ([Bauer and Swanson 2023](#)).